Dwani AI: Speech Equity for India

Making Voice Technology Accessible to Every Indian

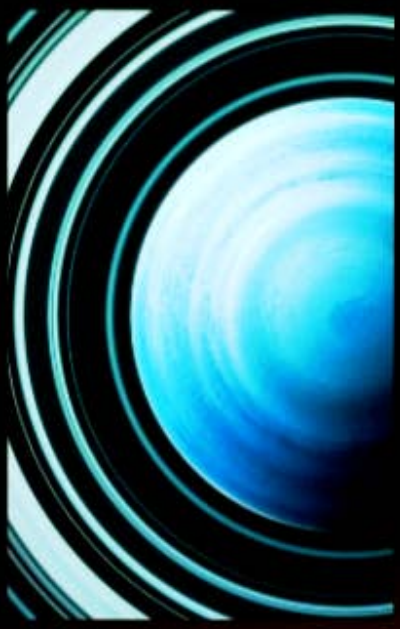
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01

The Inclusion Imperative

Dwani AI: A Universal Speech Equity Engine



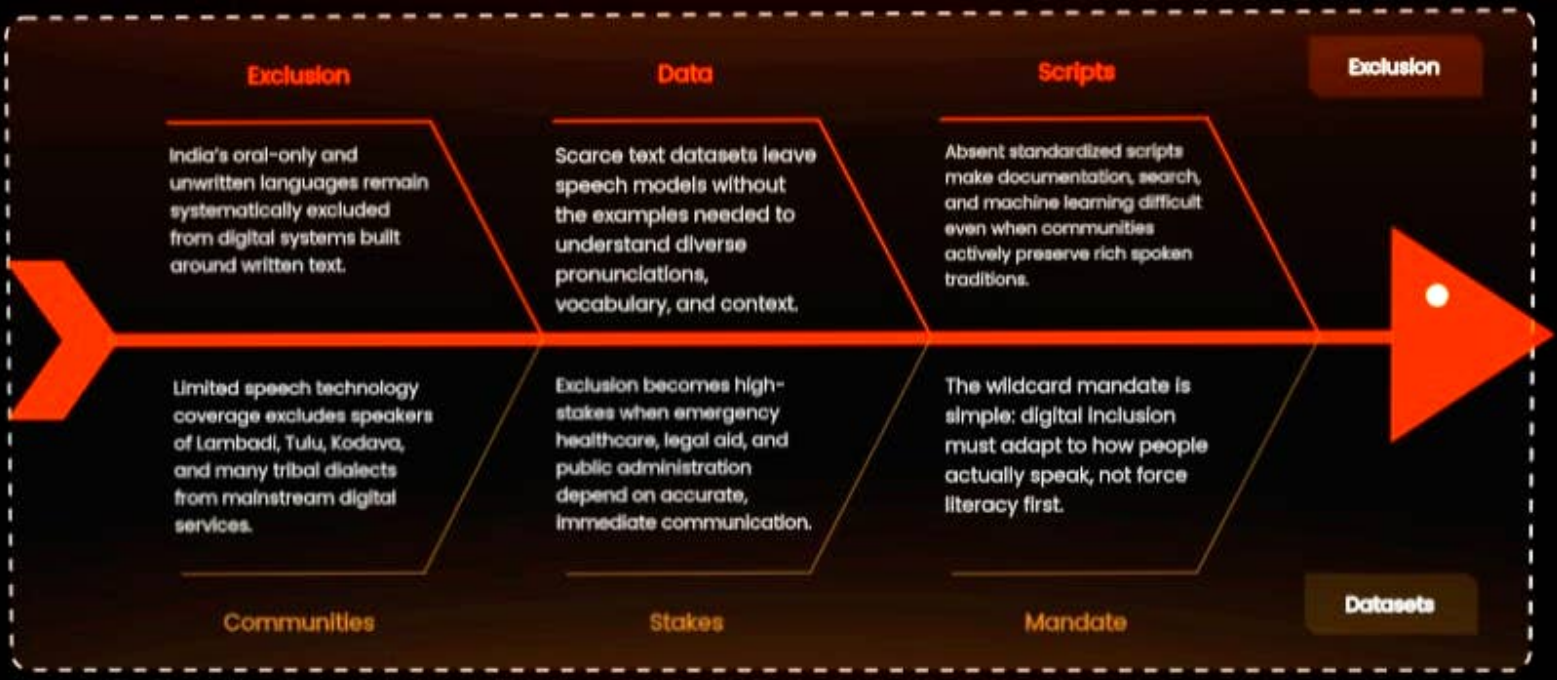
Real-Time Speech Equity for India

Dwani AI is a universal, real-time speech equity engine for India's oral-only languages, enabling diverse rural and urban communities to communicate inclusively.

Blue Brigade | Wildcard Track

Powered & Hosted on IBM LinuxONE, Blue Brigade's solution pairs resilient infrastructure with visuals of Indian communities communicating across rural and urban settings.

The Forgotten Millions



Millions face systemic exclusion as text-first technology misses languages, services, and lived realities.

02

The Scalable Equity Engine

Universal Acoustic Mapping Architecture

01

Capture Raw Vocal Streams

Capture raw vocal audio streams from diverse Indian speakers, preserving natural variation across accents, environments, devices, and speaking conditions in real-world settings.

02

Generate Universal Acoustic Vectors

Dwani AI transforms speech into universal acoustic vectors, preserving linguistic meaning for low-resource languages without requiring extensive labeled datasets or local models.

03

Deliver Low-Latency Microservices

Ultra-low-latency microservices deliver continuous uptime and rapid responses, reducing dependence on local device capacity while supporting distributed access across India.

04

Translate Through the Gateway Matrix

The Gateway Matrix maps acoustic vectors into English and regional scripts, including Kannada, enabling consistent translation across language pathways and outputs.

05

Activate High-Impact Use Cases

The architecture targets emergency response, healthcare, legal support, and public administration, turning translated speech into actionable, inclusive services for communities.

06

Compute on IBM LinuxONE

IBM LinuxONE s390x provides the secure mainframe compute core, centralizing resilient processing, governance, and protected data operations at enterprise scale.

Open Source to National Scale

Efficient Foundations

Optimized native Python network primitives, including socket operations, reduce s390x compilation overhead and establish an efficient foundation for multilingual speech services across constrained deployment environments.

Python | s390x | Vectors



Dialect Bridges

Open-source multilingual speech vectors, including Meta's MMS, connect dialect representations to mapping matrices, enabling reusable transfer across India's diverse linguistic and acoustic landscapes with greater consistency.

MMS | Dialects | Matrices

Community Archive

A crowdsourced national audio repository captures speech from unwritten Indian languages, preserving community knowledge while supplying training data for broader recognition, evaluation, and deployment at scale.

Crowdsourcing | Audio | Unwritten



Edge Access

LinuxONE microservices extend reliable speech capabilities to rural and offline edge nodes, supporting equitable, real-time access for every language community through resilient, distributed services across India.

LinuxONE | Edge | Equity

Team:Blue Brigade

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